CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 1

CANDIDATE NAME			
CLASS	2T	**INDEX NUMBER**	

BIOLOGY

8875/02

Paper 2 Core Paper

FRIDAY 26 AUGUST 2011
2 Hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper. You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

SECTION A
Answer **all** questions.

SECTION B
Answer **one** question.

At the end of the examination, fasten all work securely together.
The number of marks is given in brackets [ ] at the end of each question or part of the question.

For Examiner's Use	
Section A	**40**
1 [10]	
2 [10]	
3 [10]	
4 [10]	
Section B	**20**
5 or 6	
TOTAL	**60**

Section A

Answer **all** the questions in this section.

1 Fig. 1.1 shows two different structures, **A** and **B**, found in an animal cell.

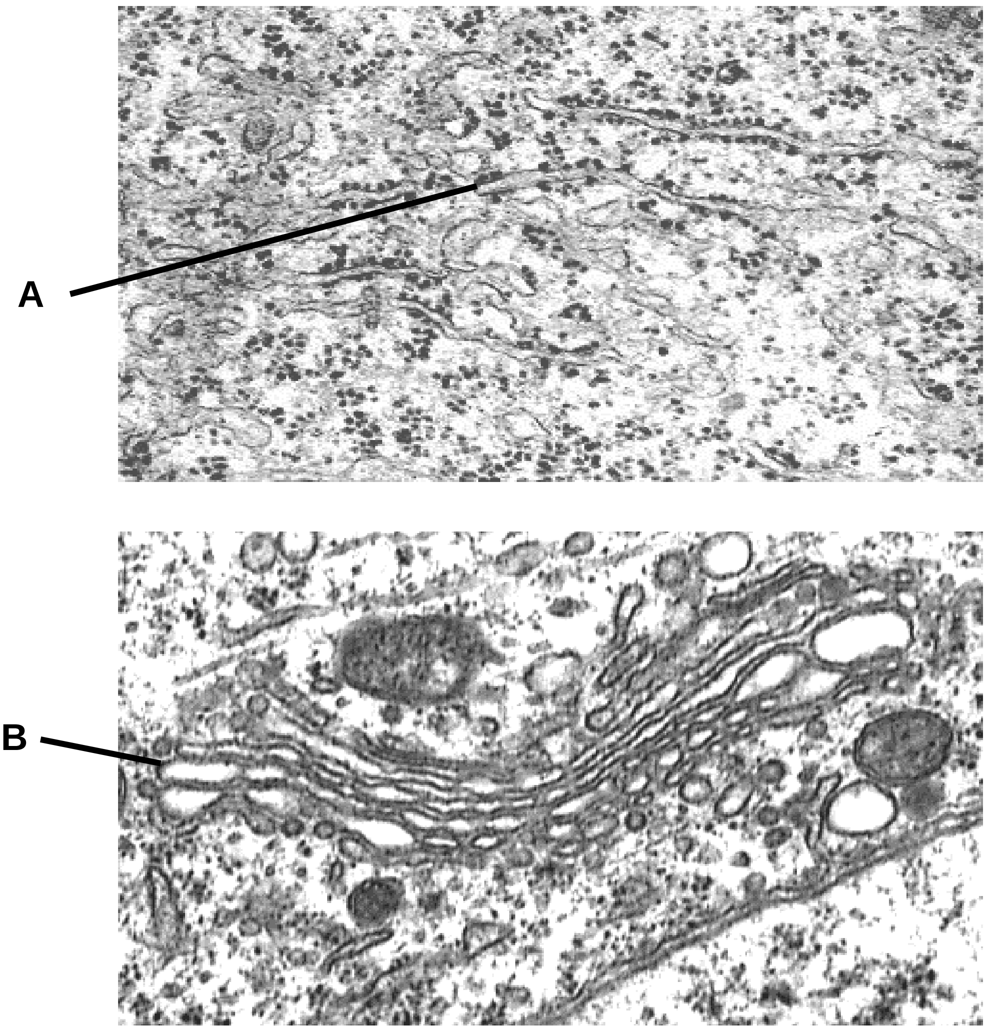

Fig. 1.1

(a) Describe two ways in which structure **A** is similar to structure **B**.

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....

....

.... [2]

Fig. 1.2 shows two worn-out mitochondria in a lysosome.

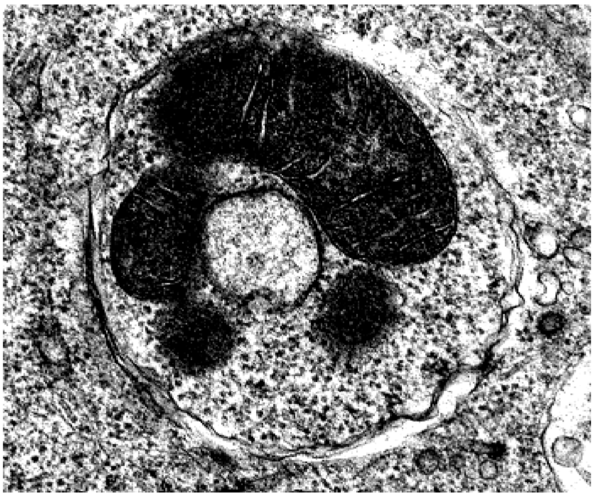

Fig. 1.2

(b) Suggest what will be the effect on the cytoplasmic enzymes when the membrane of the lysosome accidentally ruptures within the cell.

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.... [2]

One of the central tenets of biology is the Central Dogma, where DNA is used to make RNA, which in turn is used to make proteins. Ribosomes are responsible for reading genetic information transcribed from DNA and translating this information into proteins.

(c) Ribosomes are found in several locations in the eukaryotic cell. Place a tick, (✓) in the correct column to show these locations.

Location	**Ribosome present**	**Ribosome absent**
nucleus		
chloroplast		
vacuole		
mitochondria		

[2]

During the first step of translation, amino acids need to be activated and attached to its specific tRNA molecule before they can be used by the ribosomes.

Fig. 1.3 shows a tRNA and the enzyme that attaches the correct amino acid to it.

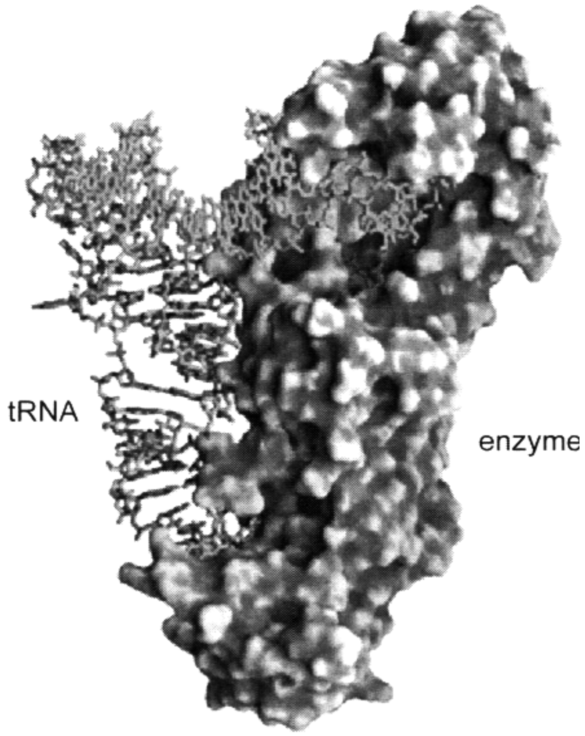

Fig. 1.3

(d) State the name of the enzyme in Fig. 1.3 and explain how the structure of the enzyme is suited for its function.

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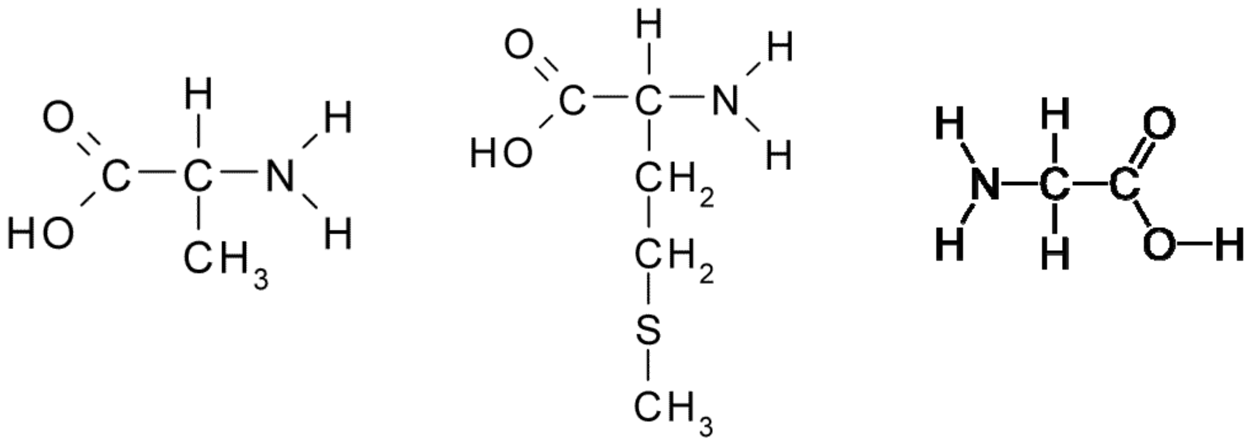
.... [2]

Proteins are chains of amino acids joined together by the action of ribosomes. Fig. 1.4 shows the structure of 3 amino acids. Methionine is the first amino acid of all proteins in eukaryotes.



Alanine Methionine Glycine

Fig. 1.4

(e) Draw the resulting tripeptide (Methionine-Glycine-Alanine) that would be formed when these 3 amino acids are joined. Draw a labelled circle around a peptide linkage. [2]

[Total: 10]

2 Fig. 2.1 shows the absorption spectrum of chorophyll a and Fig. 2.2 shows the action spectrum for photosynthesis.

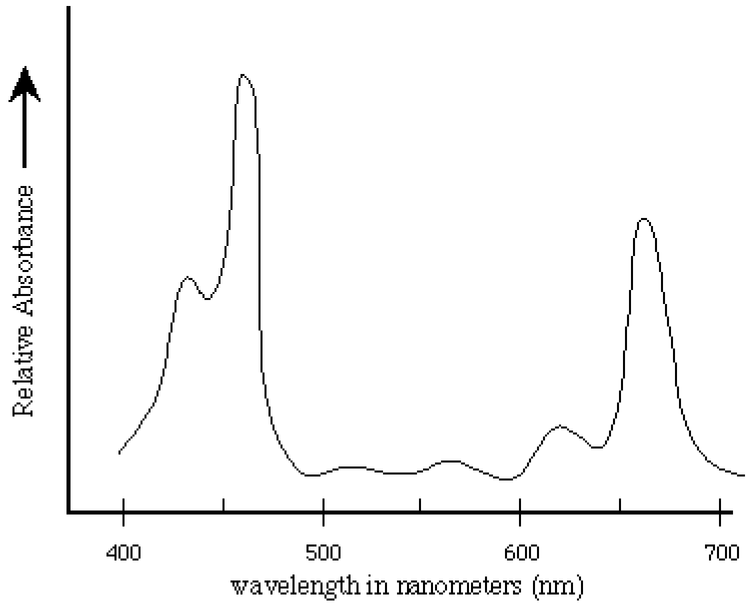


Fig. 2.1

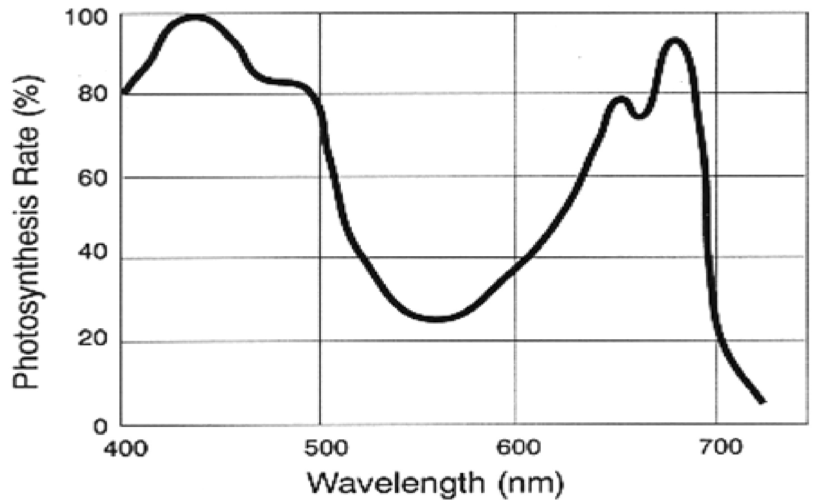


Fig. 2.2

(a) With reference to Fig. 2.1 and Fig. 2.2, explain the role of the light harvesting complex in both photosystems I and II.

[3]

(b) Distinguish the *electron transport chain* in photophosphorylation from that in oxidative phosphorylation.

Photophosphorylation	Oxidative Phosphorylation

[2]

The ATP and NADPH synthesized during photophosphorylation are utilised in the Calvin cycle for the synthesis of glucose, which is subsequently oxidised by the cell to produce ATP for its biochemical processes. Glucose may also be converted to sucrose and transported to the roots to be stored as starch.

(c) Suggest a reason why ATP generated in the chloroplast is not used directly for the cell's biochemical processes.

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____ [1]

Ribulose-1,5-bisphosphate carboxylase-oxygenase (RuBisCO) is one of the most inefficient enzymes known. Typical enzymes can process thousands of molecules in one second, while RuBisCO only fixes 3 carbon dioxide molecules per second.

In addition, RuBisCO lacks specificity, and is able to bind both oxygen and carbon dioxide.

Table 2.1 shows an experiment that measures the rate of dark reactions as amount of carbohydrates produced under different conditions.

Experiment No.	Amount of CO_2 available in stroma (%)	Amount of O_2 available in stroma (%)	pH	Amount of carbohydrates produced (Arbitrary Units)
1	10	0.002	3	0.01
2	10	0.002	8	250
3	2	0.008	3	0.01
4	2	0.008	8	15

Table 2.1

(d) With reference to Table 2.1,

(i) describe the effect of increasing pH on the amount of carbohydrates produced;

[1]

(ii) describe the effect of O_2 on the amount of carbohydrates produced at high pH.

[1]

(e) Explain the effect of a high ratio of oxygen to carbon dioxide on the amount of carbohydrates produced.

[2]

[Total: 10]

3 Warfarin is a poison used to control rat populations.

Fig. 3.1 shows changes in the proportion of rats resistant to warfarin in a particular population over a period of about 4 years. High levels of warfarin were used on this population during Year 2 but poisoning stopped at the end of this period. Rats are reproductively mature at an age of three months and can breed about every three weeks.

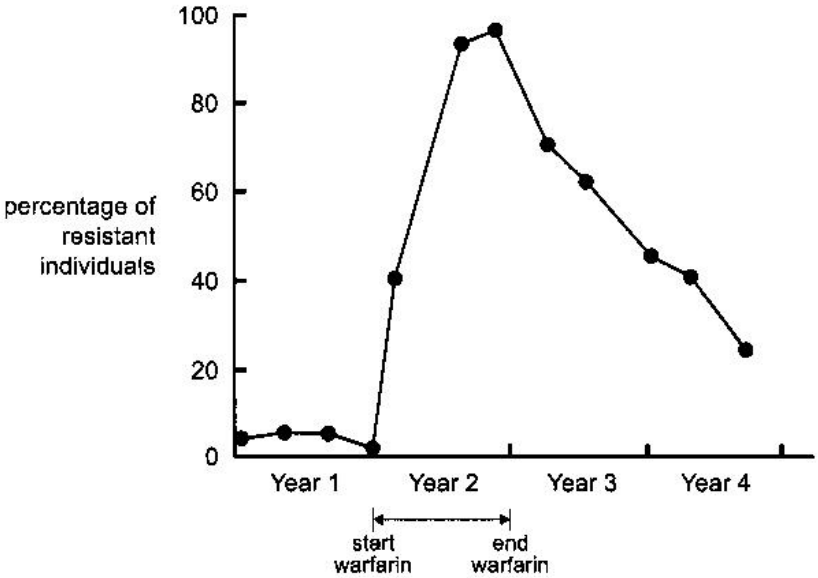

Fig. 3.1

(a) Explain the process which led to the increase in the percentage of resistant rats during Year 2.

[3]

(b) Using the data Fig. 3.1, explain what can be concluded about the selective advantage to a rat of being warfarin-resistant compared to being non-resistant in an environment without warfarin.

[2]

One population of rats, found on a remote island off the coast of South Australia, have had no genetic contact with mainland rats since they were isolated by rising sea levels at the end of the last glacial period, around 10 000 years ago.

Scientists have taken blood samples from the rats and compared the distribution of unique DNA sequences called microsatellites, which are scattered across the rats' chromosomes. These microsatellites give a measure of the population's genetic diversity, or lack of it. In this case the microsatellite data showed that the island population has low genetic diversity.

Despite the island rats' lack of genetic diversity, the population size has been maintained over many generations. In fact, the rats appear to be thriving.

(c) Suggest one reason for the rats' success despite the lack of genetic diversity within the population.

____

____ [1]

(d) Molecular techniques have been employed to study evolutionary relationships between extinct and extant organisms. Explain why DNA analysis is more advantageous than protein analysis as one of these molecular techniques.

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____ [2]

Fig. 3.2 below shows a male human karyotype.

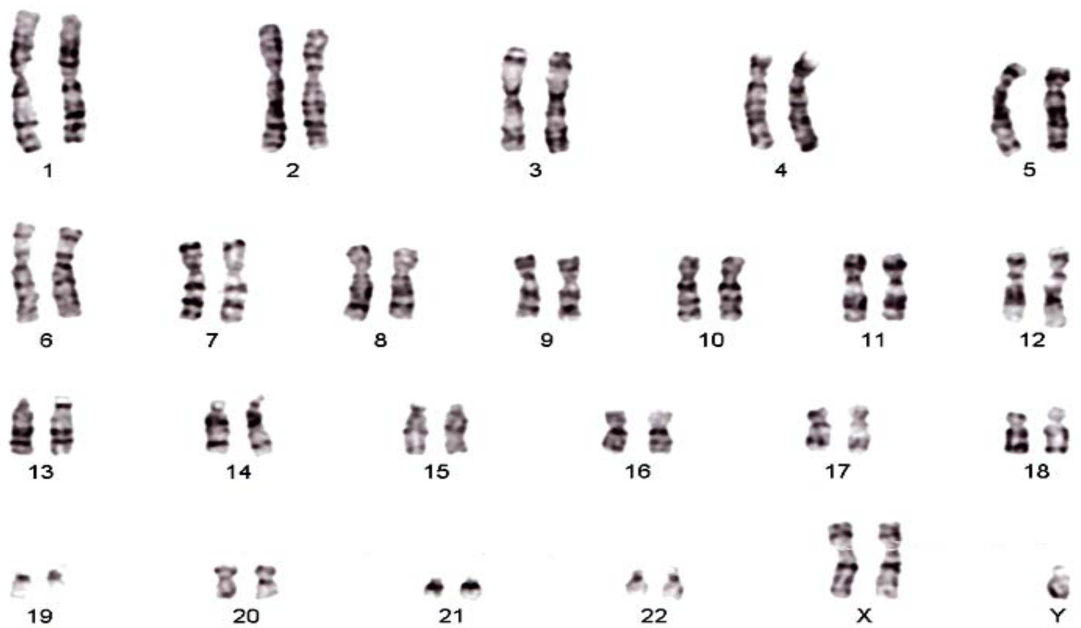

Fig. 3.2

(e) Explain what happened during meiosis I that could have produced the chromosomes shown in this karyotype.

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____ [2]

[Total: 10]

4 Factor VIII is a glycoprotein synthesized in liver cells. Many haemophiliacs, who are deficient in Factor VIII, are now treated by regular injections of genetically engineered Factor VIII.

Fig. 4.1 shows the molecular structure of Factor VIII.

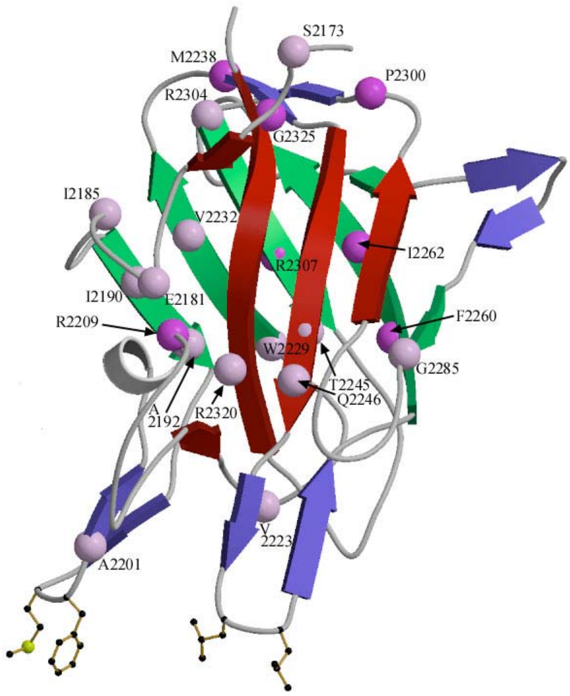

Fig 4.1

(a) Outline how the recombinant DNA encoding for Factor VIII is obtained using genetic engineering technique.

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____ [2]

(b) State one advantage of using recombinant factor VIII instead of blood derived Factor VIII.

____

____ [1]

(c) Suggest why the host cell used to produce genetically engineered Factor VIII must be a mammalian cell and not a bacterial cell.

____

____ [1]

Despite the potential benefits of GM crops such as Bt corn, there is concern over the possible environmental and agronomic impacts should the transgenes in these GM crops 'escape' and become established in both natural and agricultural ecosystems.

(d) State one problem of transgenes 'escaping' into the natural ecosystem.

____

____ [1]

Besides genetically modifying crop plants, animals have also been genetically engineered for various purposes. Salmon that have been genetically modified can grow up to 25 times larger than their wild relatives. These genetically modified salmon are sometimes termed as "super salmon".

There have been some myths about these "super salmon".

Myth 1: Transgenic salmon grow much larger than other salmon - so much so that they could gain a mating advantage or outcompete native salmon for food or space.

Myth 2: Transgenic salmon produce antifreeze proteins and excessive amounts of growth hormone.

(e) Give reasons why the two myths about "super salmon" are not true.

Myth 1:

____

____ [1]

Myth 2:

____

____ [1]

Salmon growth hormone (GH) has been inserted into plasmids and the recombinant plasmids were microinjected into eggs from very slow-growing domesticated and wild strains of Atlantic salmon. Their growths were scored, as shown in Fig. 4.2.

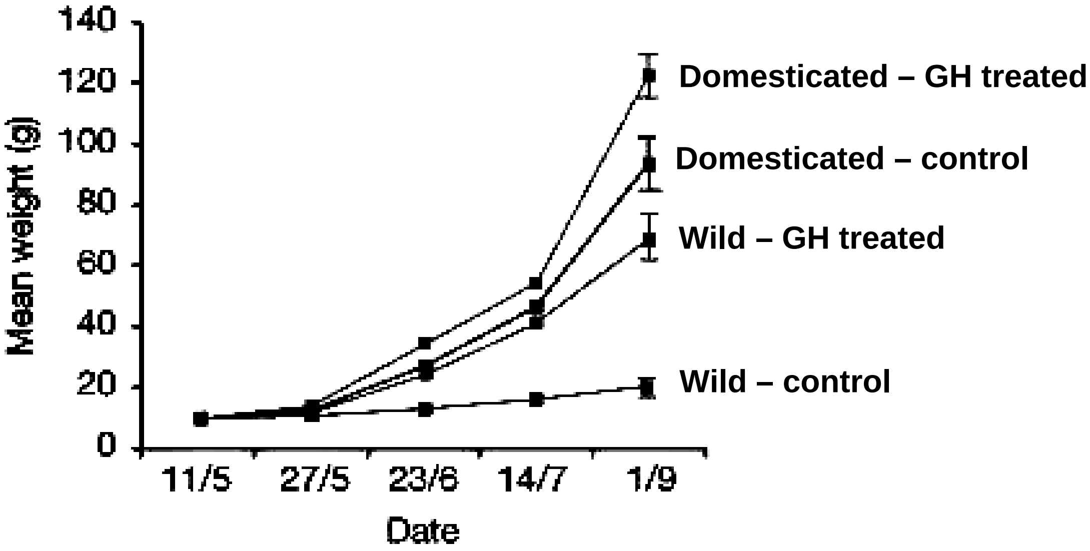

Fig. 4.2

(f) Using the information given in Fig. 4.2,

(i) describe the effect of the growth hormone on salmons;

....

....

....

.... [2]

(ii) suggest why there is a difference in weight gain between domestic-GH treated and wild-GH treated fish.

....

.... [1]

[Total: 10]

Section B

Answer **one** question.

Write your answers on separate answer paper provided.
Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.
Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 **(a)** Compare the structure of DNA and RNA molecules. [6]

(b) Describe how the information on DNA is used to synthesize a polypeptide chain of a haemoglobin molecule. [8]

(c) Using the example of sickle-cell anaemia, explain how a gene mutation may affect the phenotype of an organism. [6]

[Total: 20]

6 **(a)** Describe, with examples, the roles of proteins in membranes. [6]

(b) With reference to the analysis of DNA molecules, describe the polymerase chain reaction (PCR) and explain its advantages and limitations. [8]

(c) Describe, with reference to the unique features of stem cells, their use in the treatment of blood disorders. [6]

[Total: 20]

END OF PAPER